## 1. Identification

### Product identifier used on the label

## Luprosil®

### Recommended use of the chemical and restriction on use

Recommended use*: feed additive(s)

* The “Recommended use” identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

### Details of the supplier of the safety data sheet

Company:
BASF Mexicana S.A. de C.V.
Av. Insurgentes Sur 975
Col. CD. De Los Deportes,
C.P. 03710 Ciudad de México
MÉXICO

Telephone: +52 55 5325 2600

### Emergency telephone number
24 Hour Emergency Response Information
SETIQ: 1800-00-214-(Rep. Mexicana) or 55-59-15-88 (CDMX)
Telephone: +1-800-849-5204 or +1-833-229-1000

### Other means of identification

| | |
|---|---|
| Chemical family: | No data available. |
| Synonyms: | Propionic Acid<br>Propanoic Acid |

## 2. Hazards Identification

**According to Regulation NOM-018-STPS-2015**

### Classification of the product

| | | |
|---|---|---|
| Flam. Liq. | 3 | Flammable liquids |
| Skin Corr. | 1B | Skin corrosion |

| | | |
|---|---|---|
| Eye Dam. | 1 | Serious eye damage |
| STOT SE | 3 (irritating to respiratory system) | Specific target organ toxicity — single exposure |

### Label elements

Pictogram:

Signal Word:
Danger

Hazard Statement:

| | |
|---|---|
| H226 | Flammable liquid and vapour. |
| H335 | May cause respiratory irritation. |
| H314 | Causes severe skin burns and eye damage. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear protective gloves, protective clothing and eye protection or face protection. |
| P271 | Use only outdoors or in a well-ventilated area. |
| P210 | Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. |
| P243 | Take action to prevent static discharges. |
| P260 | Do not breathe mist or vapour or spray. |
| P241 | Use explosion-proof electrical, ventilating and lighting equipment. |
| P264 | Wash contaminated body parts thoroughly after handling. |
| P240 | Ground and bond container and receiving equipment. |
| P242 | Use non-sparking tools. |

Precautionary Statements (Response):

| | |
|---|---|
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P310 | Immediately call a POISON CENTER or physician. |
| P303 + P361 + P353 | IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. |
| P304 + P340 | IF INHALED: Remove person to fresh air and keep comfortable for breathing. |
| P301 + P330 + P331 | IF SWALLOWED: rinse mouth. Do NOT induce vomiting. |
| P363 | Wash contaminated clothing before reuse. |
| P370 + P378 | In case of fire: Use water spray, dry powder, foam or carbon dioxide for extinction. |

Precautionary Statements (Storage):

| | |
|---|---|
| P233 | Keep container tightly closed. |
| P403 + P235 | Store in a well-ventilated place. Keep cool. |
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents/container in accordance with local regulations. |

### Hazards not otherwise classified

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

## 3. Composition / Information on Ingredients

**According to Regulation NOM-018-STPS-2015**

propionic acid
CAS Number: 79-09-4
Content (W/W): >= 99.5 - <= 100.0%
Synonym: Propanoic acid; Propionic acid

Acetic acid
CAS Number: 64-19-7
Content (W/W): >= 0.0 - <= 0.2%
Synonym: Acetic acid; Glacial acetic acid

## 4. First-Aid Measures

### Description of first aid measures

**General advice:**
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

**If inhaled:**
Remove the affected individual into fresh air and keep the person calm. Assist in breathing if necessary. Immediate medical attention required.

**If on skin:**
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

**If in eyes:**
Hold eyes open and rinse slowly and gently with water for 15 to 20 minutes. Remove contact lenses, if present, after first 5 minutes, then continue rinsing. Seek medical attention.

**If swallowed:**
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

### Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

*Information on: propionic acid*
*Symptoms: Overexposure may cause:, asthma, coughing*
---

### Indication of any immediate medical attention and special treatment needed

Note to physician
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

### Extinguishing media

Suitable extinguishing media:
water spray, dry powder, foam, carbon dioxide

Unsuitable extinguishing media for safety reasons:
water jet

### Special hazards arising from the substance or mixture

Hazards during fire-fighting:
carbon oxides, nitrogen oxides
The substances/groups of substances mentioned can be released in case of fire.

### Advice for fire-fighters

Protective equipment for fire-fighting:
Wear self-contained breathing apparatus and chemical-protective clothing.

**Further information:**
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

**Impact Sensitivity:**
Remarks: Based on the chemical structure there is no shock-sensitivity.

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

Personal protection: wear a tightly closed chemical protection suit and a self-contained breathing apparatus. Wear acid-resistant boots.

### Environmental precautions

Do not empty into drains.

### Methods and material for containment and cleaning up

For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr). Dispose of absorbed material in accordance with regulations.

## 7. Handling and Storage

### Precautions for safe handling

Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift. Change clothes immediately after contamination.

Protection against fire and explosion:
Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

### Conditions for safe storage, including any incompatibilities

Segregate from alkalies and alkalizing substances.

Further information on storage conditions: Keep container tightly closed in a cool, well-ventilated place.

Storage stability:
Storage temperature: < 30 °C
Storage duration: <= 36 Months
From the data on storage duration in this safety data sheet no agreed statement regarding the warrantee of application properties can be deduced.

## 8. Exposure Controls/Personal Protection

**Components with occupational exposure limits**

| | | |
|---|---|---|
| Acetic acid | OEL, MX: | TWA value 10 ppm ; |
| | OEL, MX: | STEL value 15 ppm ; |
| propionic acid | OEL, MX: | TWA value 10 ppm ; |

**Personal protective equipment**

**Respiratory protection:**
Wear a NIOSH-certified (or equivalent) acid gas/organic vapour respirator. Do not exceed the maximum use concentration for the respirator facepiece/cartridge combination. For emergency or non-routine, high exposure situations, use a NIOSH-certified full facepiece pressure demand self-contained breathing apparatus (SCBA) or a full facepiece pressure demand supplied-air respirator (SAR) with escape provisions.

**Hand protection:**
Chemical resistant protective gloves should be worn to prevent all skin contact., Consult with glove manufacturer for testing data., Protective glove selection must be based on the user's assessment of the workplace hazards.

**Eye protection:**
Tightly fitting safety goggles (chemical goggles) and face shield.

**Body protection:**
Body protection must be chosen depending on activity and possible exposure, e.g. head protection, apron, protective boots, chemical-protection suit.

**General safety and hygiene measures:**
Avoid contact with the skin, eyes and clothing. Avoid inhalation of vapour. Avoid contact with skin and eyes. Take off immediately all contaminated clothing.

## 9. Physical and Chemical Properties

| | | |
|---|---|---|
| Form: | liquid | |
| Odour: | pungent | |
| Odour threshold: | not determined | |
| Colour: | colourless | |
| pH value: | 2.5<br>( 100 g/l, 20 °C)<br>Literature data. | |
| Melting point: | -20 °C | |
| Freezing point: | No data available. | |
| Boiling point: | 140.7 - 141.6 °C | |
| Sublimation point: | No applicable information available. | |
| Flash point: | 53 °C | (ISO 13736, closed cup) |
| Flammability: | Flammable liquid and vapour. | (derived from flash point) |
| Lower explosion limit: | 2.1 %(V)<br>( 46.9 °C)<br>The lower explosion point of the substance/mixture has been determined. The explosion point describes the temperature of a flammable liquid at which the concentration of the saturated vapour mixed with air equals the lower explosion limit. | |
| Upper explosion limit: | 12.0 %(V) | |
| Autoignition: | 485 °C | (DIN 51794) |
| SADT: | Not a substance/mixture liable to self-decomposition according to GHS. | |
| Vapour pressure: | 5 mbar<br>( 20 °C)<br>approx. 23 hPa<br>( 50 °C) | |
| Density: | 0.993 g/cm3<br>( 20 °C)<br>Literature data.<br>0.957 g/cm3<br>( 55 °C)<br>Literature data.<br>0.9990 g/cm3<br>( 15 °C)<br>0.9610 g/cm3<br>( 50 °C) | |
| Vapour density: | > 1<br>( 20 °C)<br>Heavier than air. | (estimated) |
| Partitioning coefficient n-octanol/water (log Pow): | 0.25<br>( 25 °C)<br>0.33 | (Calculation Hansch/Leo) |
| Self-ignition temperature: | Based on its structural properties the product is not classified as self-igniting. | |
| Thermal decomposition: | not determined | |

| | |
|---|---|
| Viscosity, dynamic: | 1.102 mPa.s<br>( 20 °C)<br>Literature data. |
| Viscosity, kinematic: | No applicable information available. |
| Solubility in water: | ( 20 °C)<br>miscible |
| Solubility (quantitative): | No applicable information available. |
| Solubility (qualitative): | No applicable information available. |
| Molar mass: | 74.08 g/mol |
| Evaporation rate: | Value can be approximated from Henry's Law Constant or vapor pressure. |

## 10. Stability and Reactivity

### Reactivity
No applicable information available.

Corrosion to metals:
Corrosive effects to metal are not anticipated. In the presence of water or moisture metal corrosion cannot be excluded.

Oxidizing properties:
Based on its structural properties the product is not classified as oxidizing.

| | | |
|---|---|---|
| Formation of flammable gases: | Remarks: | Forms no flammable gases in the presence of water. |

### Chemical stability
The product is chemically stable.

### Possibility of hazardous reactions
Reacts with strong alkalies. Exothermic reaction.

### Conditions to avoid
No conditions to avoid anticipated.

### Incompatible materials
bases, non-coated metals, base metals

### Hazardous decomposition products

Decomposition products:
No hazardous decomposition products known.

Thermal decomposition:
not determined

## 11. Toxicological information

**Primary routes of exposure**

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

**Acute Toxicity/Effects**

Acute toxicity
Assessment of acute toxicity: Of low toxicity after single ingestion. Virtually nontoxic by inhalation. Of low toxicity after short-term skin contact. Inhalation-risk test (IRT): No mortality within 8 hours as shown in animal studies. The inhalation of a highly saturated vapor-air mixture represents no acute hazard.

Oral
Type of value: LD50
Species: rat (male/female)
Value: 3,455 mg/kg (similar to OECD guideline 401)

Inhalation
Type of value: LC50
Species: rat (male/female)
Value: > 19.7 mg/l (OECD Guideline 403)
Exposure time: 1 h
The vapour was tested.

Type of value: LC0
Species: rat (male/female)
Value: 24.4 mg/l (IRT)
Exposure time: 8 h
The vapour was tested.
Literature data. No mortality within the stated exposition time as shown in animal studies.

Dermal
Type of value: LD50
Species: rat (female)
Value: 3,235 mg/kg (similar to OECD guideline 402)

Assessment other acute effects
Assessment of STOT single:
Causes temporary irritation of the respiratory tract.

Irritation / corrosion
Assessment of irritating effects: Corrosive! Damages skin and eyes.

Skin
Species: rabbit
Result: Corrosive.
Method: BASF-Test

Eye
Species: rabbit
Result: irreversible damage
Method: Draize test
Literature data.

Sensitization

Assessment of sensitization: Skin sensitizing effects were not observed in animal studies.

Guinea pig maximization test
Species: guinea pig
Result: Non-sensitizing.
Method: similar to OECD guideline 406
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aspiration Hazard
No aspiration hazard expected.

**Chronic Toxicity/Effects**

Repeated dose toxicity
Assessment of repeated dose toxicity: No substance-specific organtoxicity was observed after repeated administration to animals. After repeated administration the prominent effect is the induction of corrosion.

Genetic toxicity
Assessment of mutagenicity: The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture. The substance was not mutagenic in studies with mammals. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Carcinogenicity
Assessment of carcinogenicity: In long-term animal studies in which the substance was given in high concentrations by feed, a carcinogenic effect was not observed.

Reproductive toxicity
Assessment of reproduction toxicity: No data available. Study scientifically not justified.

Teratogenicity
Assessment of teratogenicity: No indications of a developmental toxic / teratogenic effect were seen in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

## 12. Ecological Information

### Toxicity

Aquatic toxicity
Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish
LC50 (96 h) > 10,000 mg/l, Leuciscus idus (DIN 38412 Part 15, static)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic invertebrates
EC50 (48 h) > 500 mg/l, Daphnia magna (Directive 84/449/EEC, C.2, static)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic plants
EC50 (72 h) > 500 mg/l (biomass), Scenedesmus subspicatus (OECD Guideline 201, static)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Chronic toxicity to fish
Study scientifically not justified.

Chronic toxicity to aquatic invertebrates
Study scientifically not justified.

Assessment of terrestrial toxicity
Toxic effects have been observed in studies with terrestric plants.

Soil living organisms

Toxicity to soil dwelling organisms:
No data available.

Toxicity to terrestrial plants

EC50 (3 d) 125.8 mg/l 188.7 mg/kg, Lactuca sativa
Literature data.

Other terrestrial non-mammals
No data available.

### Microorganisms/Effect on activated sludge

Toxicity to microorganisms
DIN EN ISO 8192 aquatic
activated sludge, domestic/EC20 (30 min): 500 - 1,040 mg/l
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

### Persistence and degradability

Assessment biodegradation and elimination (H2O)
Readily biodegradable (according to OECD criteria). Literature data.

Elimination information

approx. 74 % BOD of the ThOD (30 d) (other) (aerobic, activated sludge, domestic)

Assessment of stability in water
According to structural properties, hydrolysis is not expected/probable.

Information on Stability in Water (Hydrolysis)
The product has not been tested. The statement has been derived from the structure of the product.

### Bioaccumulative potential

Assessment bioaccumulation potential

Significant accumulation in organisms is not to be expected.

Bioaccumulation potential
Accumulation in organisms is not to be expected.

### Mobility in soil

Assessment transport between environmental compartments
The substance will not evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is not expected.

### Additional information

Sum parameter

Chemical oxygen demand (COD): 1,520 mg/g

Biochemical oxygen demand (BOD) Incubation period 5 d: 1,300 mg/g

## 13. Disposal considerations

**Waste disposal of substance:**
Incinerate in suitable incineration plant, observing local authority regulations.

**Container disposal:**
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

## 14. Transport Information

**Land transport**
TDG

| | |
|---|---|
| Hazard class: | 8 |
| Packing group: | II |
| ID number: | UN 3463 |
| Hazard label: | 8, 3 |
| Proper shipping name: | PROPIONIC ACID |

**Sea transport**
IMDG

| | |
|---|---|
| Hazard class: | 8 |
| Packing group: | II |
| ID number: | UN 3463 |
| Hazard label: | 8, 3 |
| Marine pollutant: | NO |
| Proper shipping name: | PROPIONIC ACID |

**Air transport**
IATA/ICAO

| | |
|---|---|
| Hazard class: | 8 |
| Packing group: | II |
| ID number: | UN 3463 |
| Hazard label: | 8, 3 |

Proper shipping name: PROPIONIC ACID

## 15. Regulatory Information

**Federal Regulations**

Not applicable

**NFPA Hazard codes:**
Health: 3 Fire: 2 Reactivity: 0 Special:

**Assessment of the hazard classes according to UN GHS criteria (most recent version):**

| | | |
|---|---|---|
| Skin Corr./Irrit. | 1B | Skin corrosion/irritation |
| Acute Tox. | 5 (oral) | Acute toxicity |
| Flam. Liq. | 3 | Flammable liquids |
| Eye Dam./Irrit. | 1 | Serious eye damage/eye irritation |
| Acute Tox. | 5 (dermal) | Acute toxicity |
| STOT SE | 3 (irritating to respiratory system) | Specific target organ toxicity — single exposure |

## 16. Other Information

**SDS Prepared by:**
BASF NA Product Regulations
SDS Prepared on: 2024/12/04

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This information is considered accurate but is not exhaustive and shall only be used as a guideline based on current knowledge of the chemical substance or mixture. Safety precautions suitable for the product must be applied.

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